UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 011 · Lagrange's method of undetermined multipliers · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011·5	Deck 011, slide 5 (rendered .pptx)
T011a 00:01:08	Transcript "Lagrange's method of undetermined coefficient" (theory lecture), segment start
T011b 00:05:03	Transcript "Problems on Lagrange's method of undetermined multipliers"
CW U1 P33	Unit 1 class-work problem 33
Extra Q57	Extra question list "lagrange-ques", question 57
ESA Jul-23 1c	Past paper: Jul 2023, question 1.c

Contents

1. 1 · Lagrange's method: the auxiliary function and the procedure
2. 2 · Geometric problems: inscribed solids, nearest and farthest points
3. 3 · Applied problems: boxes, splitting a number, temperature
4. Past-paper questions

1 Lagrange's method: the auxiliary function and the procedure

Definitions

Constraint. A condition $\phi(x, y, z) = 0$ that the variables must satisfy (walk only on the surface of a sphere, a fixed surface area, a fixed sum). S011·3 · T011a 00:00:36

Lagrange multiplier. Suppose we need to minimize or maximize a function $f(x, y, z)$ subject to the constraint $\phi(x, y, z)$; then we can introduce an additional unknown constant λ, known as Lagrange's multiplier, to ease the process of finding the extrema. S011·3 · T011a 00:00:53

Auxiliary function. $F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$: the given function plus λ times its constraint (also called Lagrange's function). S011·3 · T011a 00:01:08 · T011b 00:02:06

Statements in the class material

> *"Working procedure. Step 1: Form the auxiliary equation* $F(x,y,z) = f(x,y,z) + \lambda\phi(x,y,z)$ *... (1). Step 2: Partially differentiate* F *in (1) w.r.t.* x, y, z *respectively. Step 3: Solve the four equations* $F_x = 0$, $F_y = 0$, $F_z = 0$ *and the constraint for the Lagrange multiplier* λ *and stationary values* x, y, z*."* S011·3

> *"Advantages: 1. The stationary points of* $f(x,y,z)$ *can be determined without determining* x, y, z *explicitly. 2. This method can be extended to function of several variables subject to many constraints. Disadvantages: 1. Nature of the stationary points can not be determined. 2. Equations* $F_x = 0$, $F_y = 0$, $F_z = 0$ *are only necessary conditions."* S011·4 · T011a 00:03:04

Key points

1. The method avoids elimination: in class 010 we had to substitute $z^2 = 1 - x^2 - y^2$ for the temperature problem; with Lagrange we differentiate F directly. ESA ★★ T011a 00:01:18 · T011a 00:03:17
2. Four unknowns x, y, z, λ need four equations: $F_x = 0$, $F_y = 0$, $F_z = 0$ and the constraint $\phi = 0$. ESA ★★★ T011a 00:02:30
3. λ is "undetermined": we may find it, but it does not appear in the answer. ESA ★ T011a 00:02:04
4. The method gives only stationary points; to verify maximum or minimum by $rt - s^2$ we would have to go back to the previous method. In problems the question itself says which (maximum volume, closest point). ESA ★★ S011·4 · T011a 00:04:05 · T011a 00:14:01
5. The most important step is to identify the function f and the constraint ϕ correctly (Figure F1.2). ESA ★★★ T011a 00:09:15 · T011b 00:01:46
6. Geometrically, at a constrained extremum a level curve (surface) of f just touches the constraint, so ∇f and $\nabla\phi$ are parallel: $\nabla f = -\lambda\nabla\phi$ (Figure G1.1). ESA ★ ADDED FOR CLARITY S011·7

Why?

Why does $F_x = F_y = F_z = 0$ find the constrained extrema?
At a constrained extremum, f cannot increase along any direction that stays on the surface $\phi = 0$, so ∇f is perpendicular to the surface, i.e. parallel to $\nabla\phi$: $\nabla f + \lambda\nabla\phi = 0$, which is exactly $F_x = F_y = F_z = 0$.
ADDED FOR CLARITY S011·3 · S011·7

Exam traps

Exam trap. Do not forget the constraint as the fourth equation; three equations cannot fix four unknowns. T011a 00:02:30

Exam trap. Dividing by x, y, z or λ assumes it is not zero; check those cases separately (they usually give zero volume or an impossible point). ADDED FOR CLARITY S011·9 · T011b 00:03:40

2 Geometric problems: inscribed solids, nearest and farthest points

Definitions

Continued product. The product obtained by multiplying together in succession: for parts x, y, z, the product of the first, the square of the second and the cube of the third is xy^2z^3. S011·12 · T011b 00:05:23

Key points

1. For a box inscribed in a sphere or ellipsoid centred at the origin, take the corner (x, y, z) in the first octant: edges $2x, 2y, 2z$, volume $8xyz$. Using xyz instead gives the same point. ESA ★★★ S011·5 · T011a 00:06:08 · T011a 00:06:53
2. To find the nearest point, imagine a small sphere centred at the origin expanding like a soap bubble until it just touches the surface (Figure G2.1). ESA ★ S011·7
3. Reject solutions that are impossible: on $x^2 - z^2 = 1$, $x = 0$ would need $z^2 = -1$. ESA ★★ S011·9 · T011a 00:12:26
4. When there are two stationary values (W2.4: distances 1 and 2), the larger is the maximum and the smaller the minimum on a closed curve. ESA ★★★ ADDED FOR CLARITY ESA Jul-23 1c

Why?

Why minimise $x^2 + y^2 + z^2$ instead of the distance?
$\sqrt{u}$ increases with u, so both have their extrema at the same points, and the square has no square root to differentiate.

ADDED FOR CLARITY S011·8 · T011a 00:10:08

Exam traps

Exam trap. The box in the sphere has edges $2x, 2y, 2z$, so $V = 8xyz$: its largest volume is $\frac{8r^3}{3\sqrt{3}}$, not $\frac{r^3}{3\sqrt{3}}$. ADDED FOR CLARITY S011·5

Exam trap. Slide 011·6's third line should read $8xyz + 2z^2\lambda = 0$ (printed with y^2).

ADDED FOR CLARITY S011·6

3 Applied problems: boxes, splitting a number, temperature

Key points

1. Open box (no top): surface $xy + 2yz + 2zx$; closed box: $2(xy + yz + zx)$. ESA ★★ S011·10 · T011b 00:01:09
2. Dividing a number N into parts x, y, z: constraint $x + y + z = N$; the function is the product asked for. ESA ★ S011·12 · T011b 00:05:57
3. The temperature problem of class 010 gives the same 50 by Lagrange's method with no elimination (W3.3). ESA ★★★ T011a 00:14:33 · ESA Jun-24 1d

Exam traps

Exam trap. Open-top box: $x = y = 2z$ (height half the base edge), not a cube. S011·11 · CW U1 P34

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Lagrange's method is asked in five of them, always as a 7-mark Q1 part, with four distinct problems: Jul 2023 Q1.c (maximum and minimum distances from the origin to $5x^2 + 6xy + 5y^2 - 8 = 0$), May 2024 Q1.c (the rectangular solid of maximum volume in a sphere is a cube; also slides 011·5–6), Jan 2024 sets 3 and 4 Q1.a (largest parallelepiped in an ellipsoid) and June 2024 Q1.d (highest temperature $400xyz^2$ on the unit sphere; also class 010 by elimination). Three of the four also appear in the extra question list (Q57, Q59, Q61). Dec 2023 and Jan 2024 sets 1–2 ask free extrema (class 010) instead.

2 · Geometric problems: inscribed solids, nearest and farthest points

ESA Jul 2023 (7 marks) UE22MA141A Q1.c · solved in W2.4

Find the maximum and minimum distances from the origin to the curve $5x^2 + 6xy + 5y^2 - 8 = 0$.

ESA May 2024 (7 marks) UE22MA141A Q1.c · solved in W2.1

Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube by using Lagrange's method of multipliers.

ESA Jan 2024 (7 marks) UE23MA141A set 3, Q1.a · solved in W2.2

Use the method of Lagrange's multipliers to find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

ESA Jan 2024 (7 marks) UE23MA141A set 4, Q1.a · solved in W2.2

The same question as set 3 Q1.a (largest parallelepiped in the ellipsoid).

3 · Applied problems: boxes, splitting a number, temperature

ESA June 2024 (7 marks) UE23MA141A Q1.d · solved in W3.3

The temperature T at any point (x, y, z) in space is $400xyz^2$. Using Lagrange's method of multipliers, find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.